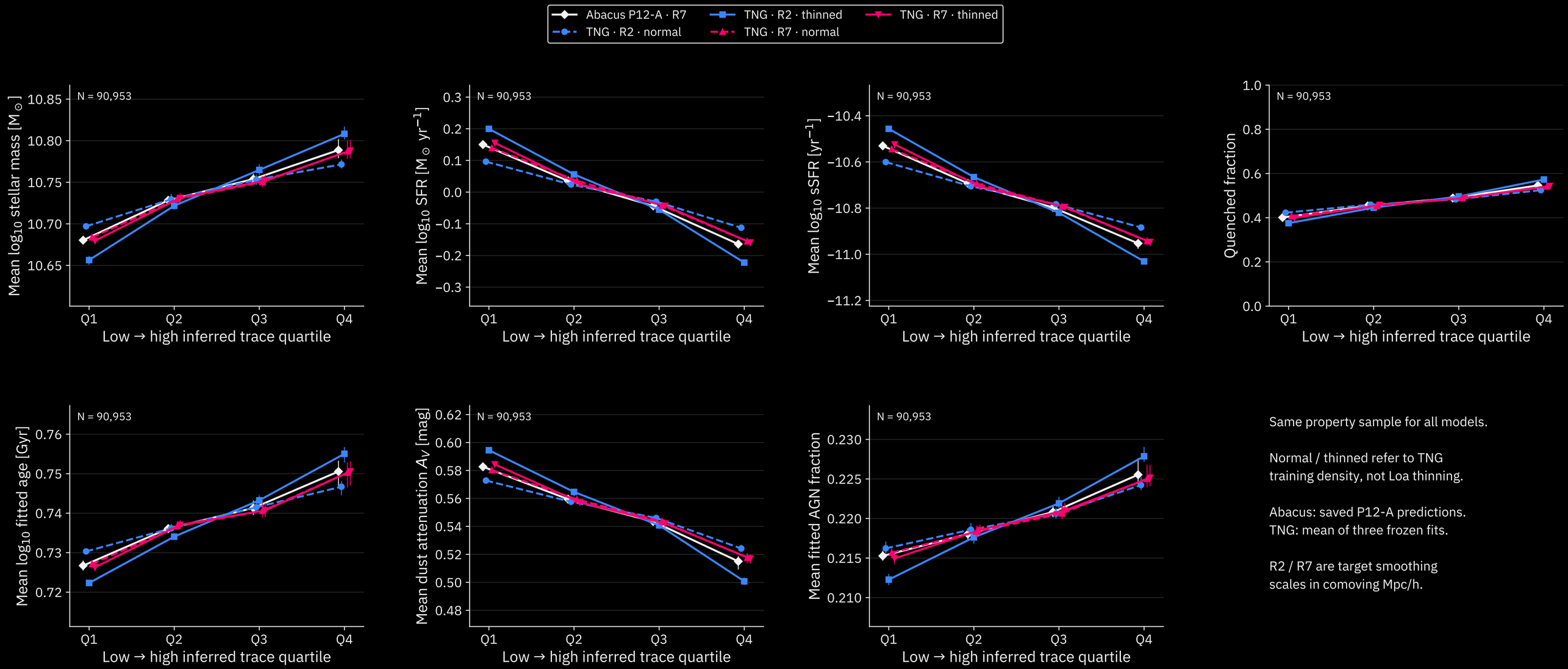
Loa properties versus inferred environment rank

Same DESI Loa galaxies · 0.20 ≤ z < 0.30 · CIGALE 15-band properties



Same property sample for all models.

Normal / thinned refer to TNG training density, not Loa thinning.

Abacus: saved P12-A predictions.
TNG: mean of three frozen fits.

R2 / R7 are target smoothing scales in comoving Mpc/h.

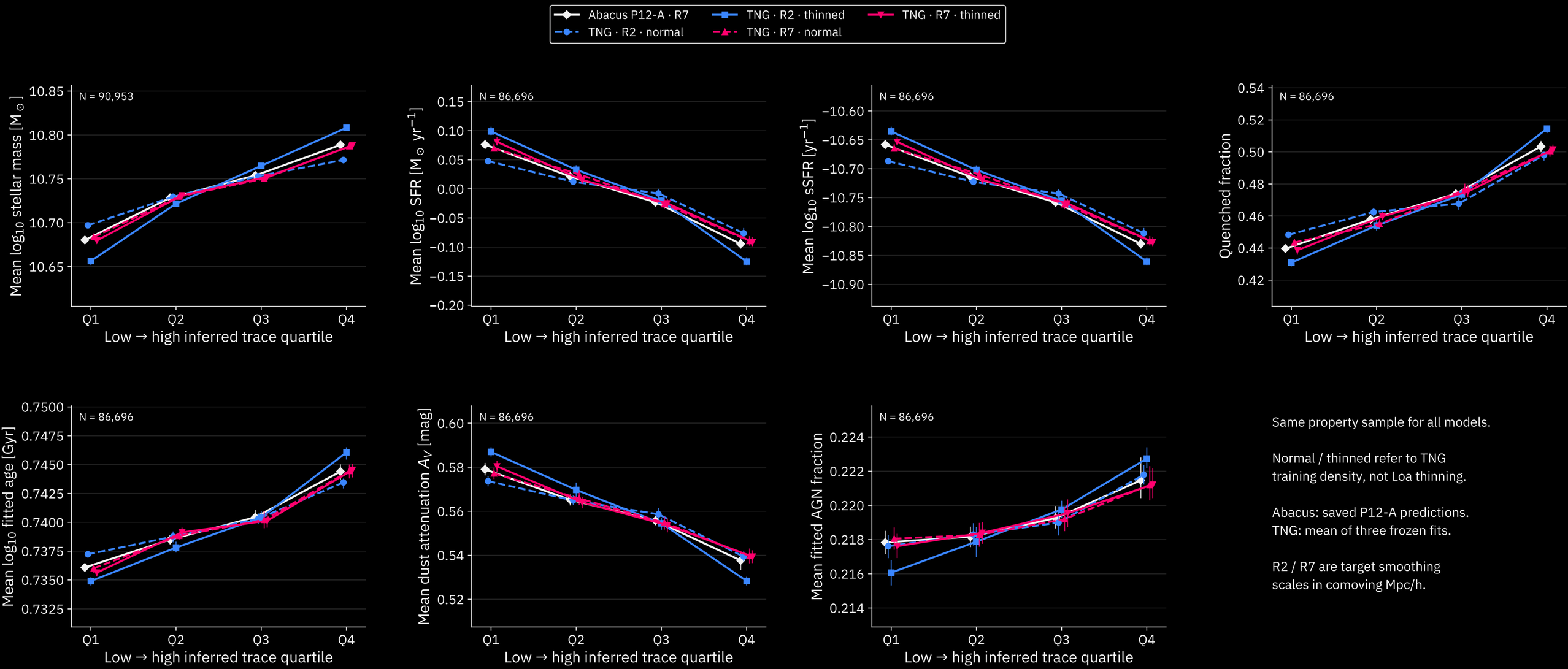
Raw selected-sample means; stellar-mass and redshift mixtures are not balanced.

Each model defines its own trace quartiles within Δz = 0.01; equal rank is not equal physical environment.

Bars: 16–84% paired sky-block bootstrap · zero-shot transfer; inferred environments are not calibrated observational truth

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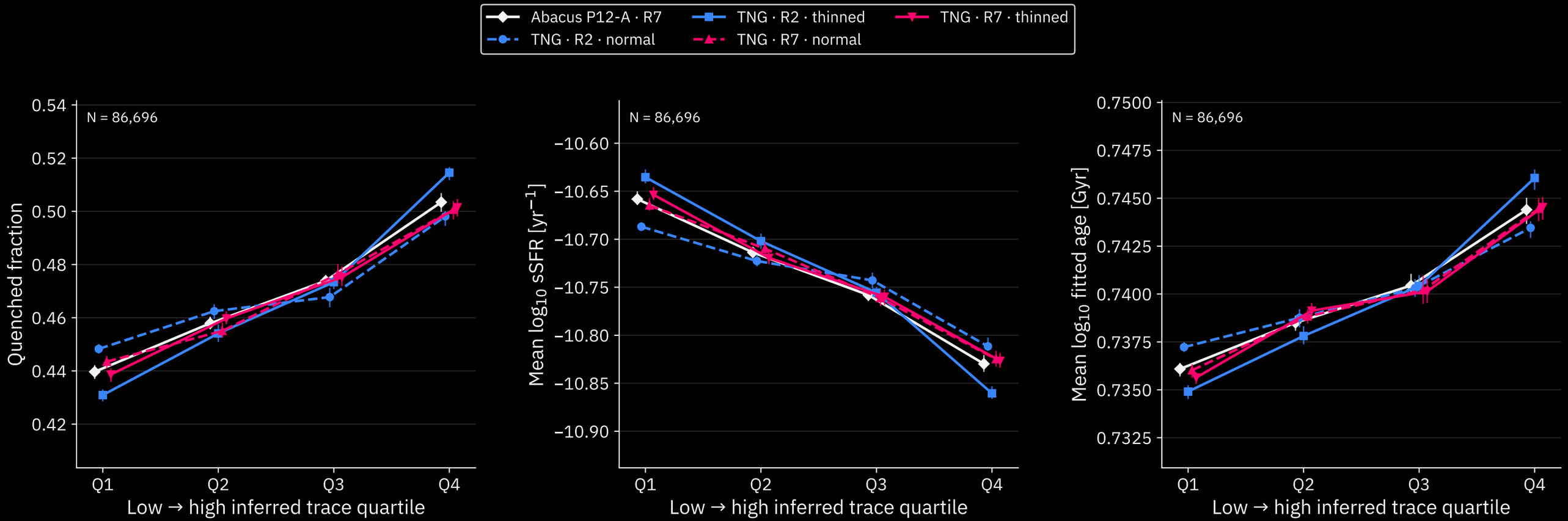
Common mass + redshift mix; stellar-mass panel controls redshift only.

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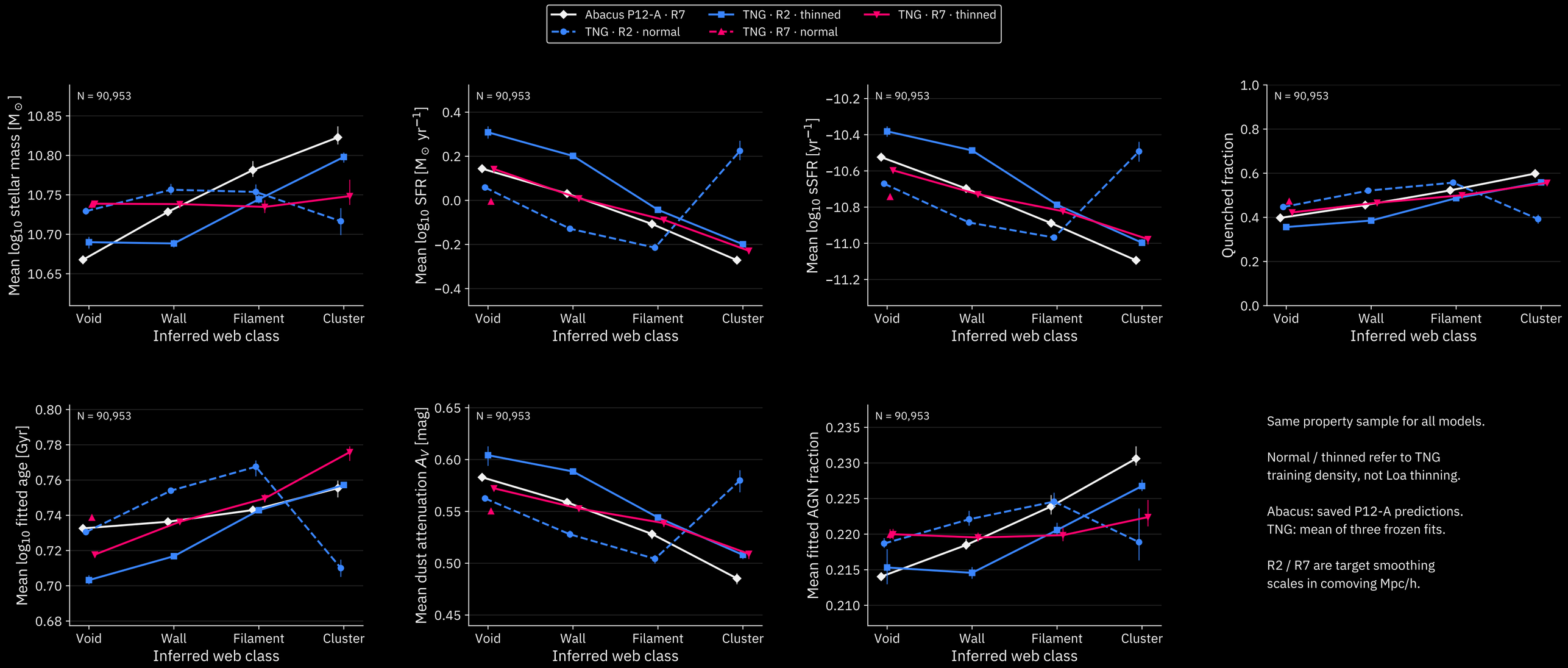
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Loa properties versus inferred web class

Same DESI Loa galaxies · 0.20 ≤ z < 0.30 · CIGALE 15-band properties



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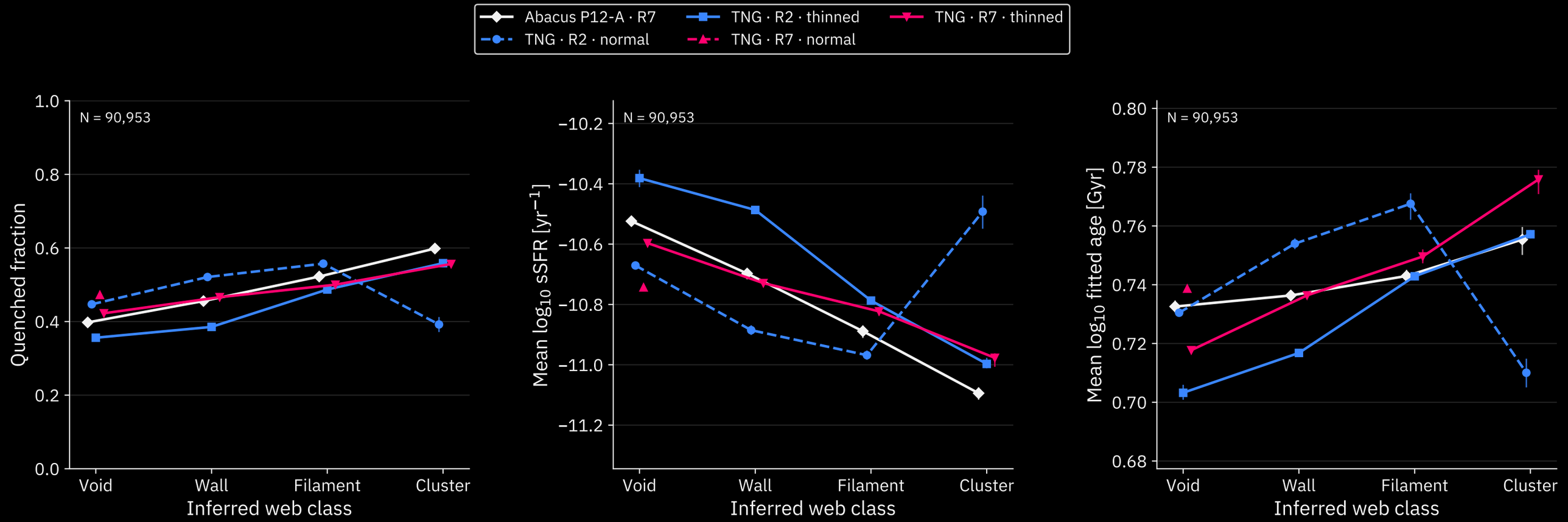
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Class gaps indicate empty groups; Abacus classes use posterior-mean eigenvalues, not posterior probabilities.

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Same DESI Loa galaxies · 0.20 ≤ z < 0.30 · CIGALE 15-band properties



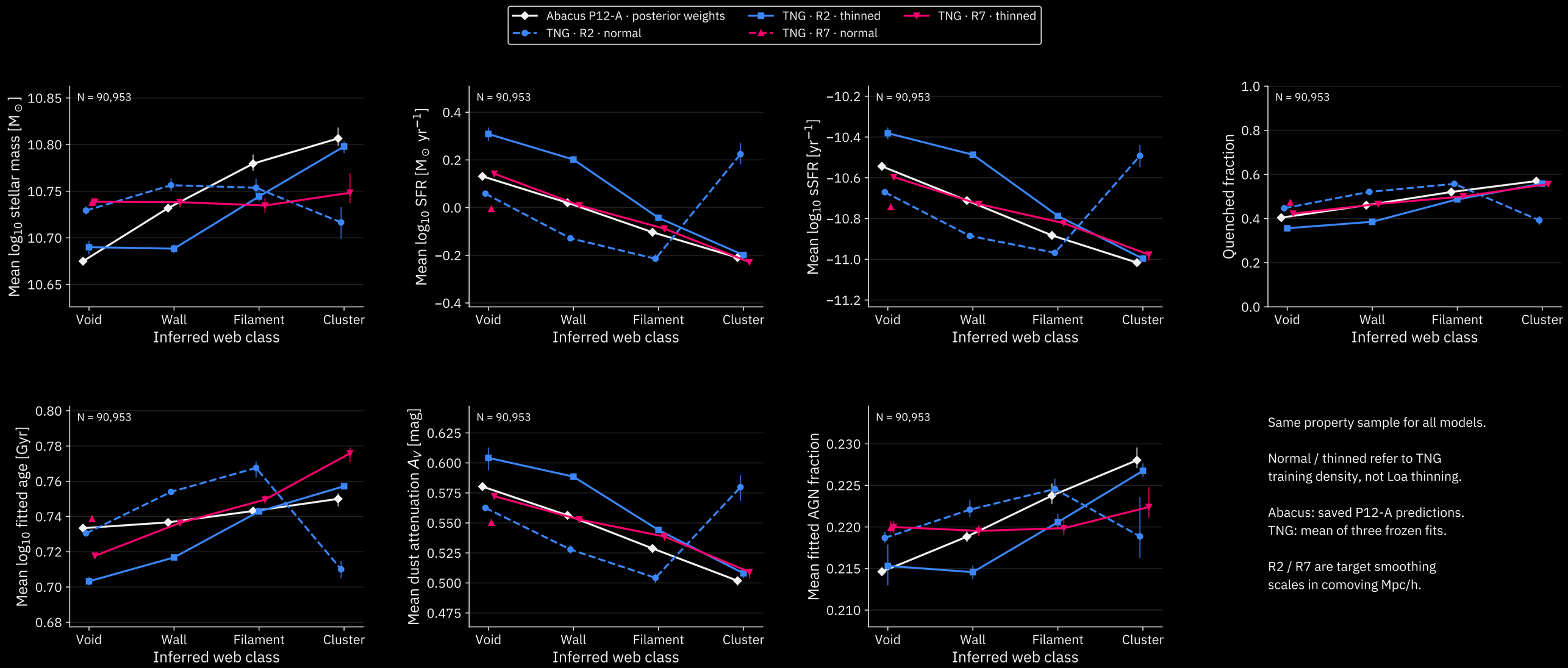
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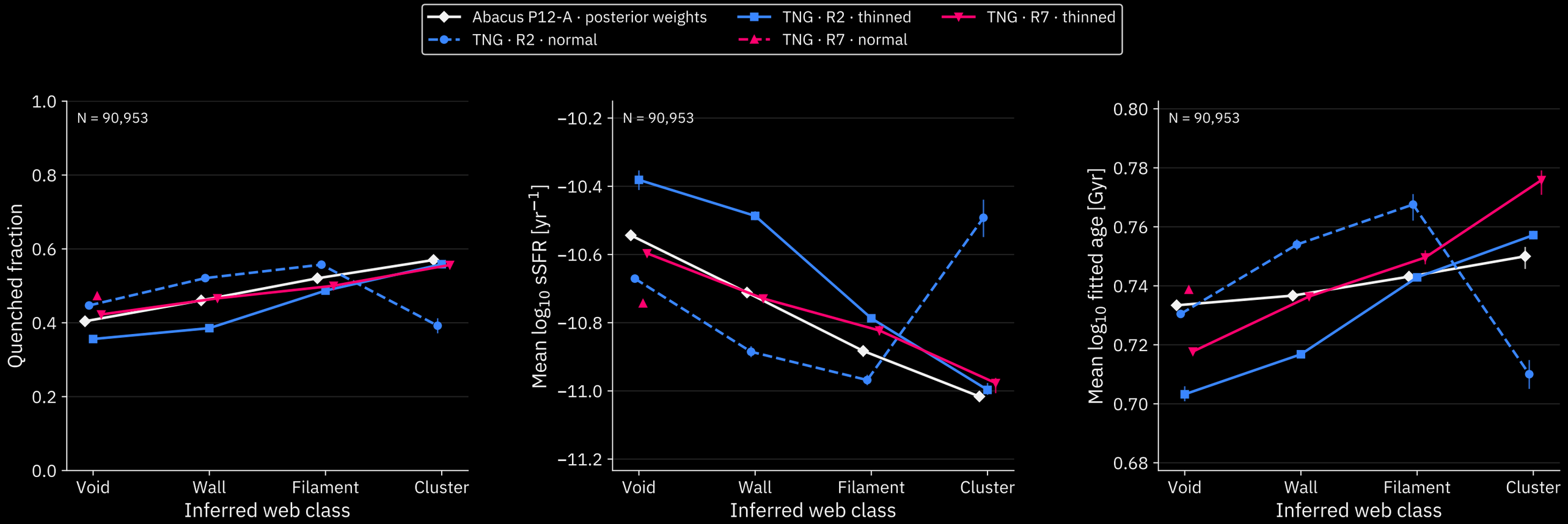
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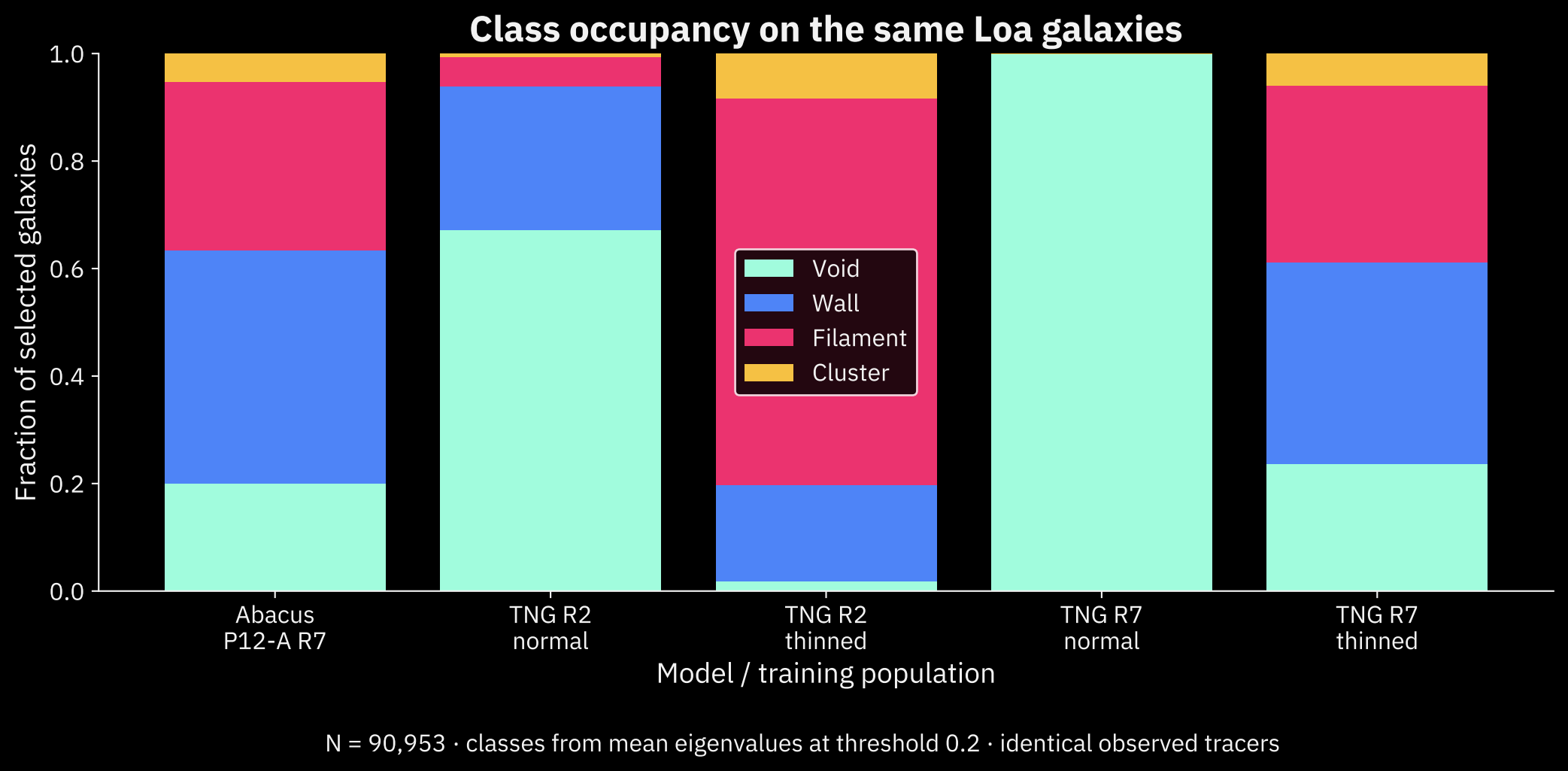
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Class occupancy on the same Loa galaxies
Fraction of selected galaxies
1.0
0.8
0.6
0.4
0.2
0.0
Void
Wall
Filament
Cluster
Abacus
P12-A R7
TNG R2
normal
TNG R2
thinned
TNG R7
normal
TNG R7
thinned
Model / training population
N = 90,953 · classes from mean eigenvalues at threshold 0.2 · identical observed tracers